🔑 **Answer Key — Teacher Copy · Lesson 3-7**

MSTAR PRACTICE WORKSHEET

Lesson 3-7: Ratio Reasoning: Convert Measurements Between Systems

Grade 6 Mathematics · Standard 6.AT.3

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: A

1 kilometer ≈ 0.6 mile. About how many miles is 5 kilometers?

- **A** 3 miles
- **B** 5.6 miles
- **C** 8 miles
- **D** 30 miles

Scale the conversion factor by 5: 0.6 × 5 = 3 miles.

Question 2 SELECTED RESPONSE — CORRECT: A

1 pound ≈ 0.45 kilogram. About how many kilograms is 10 pounds?

- **A** 4.5 kg
- **B** 10.45 kg
- **C** 22 kg
- **D** 45 kg

Scale by 10: 0.45 × 10 = 4.5 kilograms.

Question 3 SELECTED RESPONSE — CORRECT: D

Which pair of units comes from two DIFFERENT measurement systems?

A Inches and feet

B Cups and quarts

C Grams and kilograms

D Miles and kilometers

Miles are customary and kilometers are metric, so comparing them requires a between-systems conversion. The other pairs stay inside one system.

Question 4 SELECTED RESPONSE — CORRECT: B

A sign in Mexico gives a distance of 80 kilometers. Using 1 km ≈ 0.6 mi, about how far is that in miles?

A About 8 miles

B About 48 miles

C About 80 miles

D About 133 miles

0.6 × 80 = 48 miles. Because a kilometer is shorter than a mile, the number of miles is smaller than the number of kilometers.

Question 5 SELECTED RESPONSE — CORRECT: B

A dog weighs 31 pounds. Flea medicine comes in packages for 4–10 kg, 10–25 kg, and 25–40 kg. Which package is correct?

A The 4–10 kg package

B The 10–25 kg package

C The 25–40 kg package

D None — 31 is above every range

31 pounds ≈ 0.45 × 31 ≈ 14 kilograms, which falls inside the 10–25 kilogram package.

Question 6 SELECTED RESPONSE — CORRECT: C

1 inch ≈ 2.54 cm. About how many centimeters is 4 inches?

A 1.57 cm

B 6.54 cm

C 10.16 cm

D 254 cm

Scale the conversion factor by 4: 2.54 × 4 = 10.16 centimeters.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: D

A backpack weighs 5 kilograms. Using 1 kilogram ≈ 2.2 pounds, about how much does it weigh in pounds?

A 2.27 pounds

B 7.2 pounds

C 10 pounds

D 11 pounds

5 kilograms × 2.2 pounds per kilogram = 11 pounds.

- **A:** That divides 5 by 2.2. The conversion multiplies, since each kilogram IS 2.2 pounds.
- **B:** That adds 5 and 2.2. Use the conversion factor as a multiplier: 5 x 2.2.
- **C:** That uses 2 pounds per kilogram instead of 2.2. The extra 0.2 per kilogram adds a full pound over five.

Part B — correct answer: C

Why is the symbol ≈ used in the conversion 1 kilogram ≈ 2.2 pounds?

A The symbol means the units are unrelated.

B Pounds and kilograms measure different quantities.

C The conversion between these systems is approximate, not exact.

D The conversion only works for small numbers.

1 kilogram is about 2.2046 pounds. Between-system conversions are usually rounded, so results are approximations rather than exact values.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, C, E

Select ALL statements that are true about converting measurements between systems:

- **A** A conversion ratio connects a unit in one system to a unit in another.
- **B** Converting to a smaller unit produces a larger number.
- **C** Between-system conversions are often approximate.
- **D** The two measurements describe different physical amounts after converting.
- **E** A ratio table can organize a between-system conversion.

A, B, C, and E are true. D is false — converting changes the unit and the number, but the physical amount being measured stays exactly the same.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

The Golden Gate Bridge can hold about 35,924,000 pounds. An average midsize car weighs about 1,524 kilograms. Using 1 kilogram ≈ 2.2 pounds, about how many midsize cars could be on the bridge at one time? Explain each step, and say why your answer is an estimate.

Model answer: First convert one car's mass into pounds so both amounts share a unit: 1,524 × 2.2 ≈ 3,353 pounds per car. Then divide the bridge's capacity by the weight of one car: 35,924,000 ÷ 3,353 ≈ 10,700 cars. The answer is an estimate for two reasons: the conversion factor 2.2 is rounded (1 kilogram is about 2.2046 pounds), and 1,524 kilograms is itself an average car mass, not an exact one. Reporting 'about 10,700 cars' is honest; reporting 10,714.6 cars would claim precision the inputs never had.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.